

Hudson Hall

704-619-4872 | hch28@stanford.edu | <https://hudson-hall-portfolio.vercel.app/>

EDUCATION

Stanford University

Stanford, CA

Bachelor of Science in Engineering Physics, GPA: 3.731

Sept. 2024 – June 2028

Organizations

Executive Member of the Sigma Chi Alpha Omega Chapter

EXPERIENCE

Student Researcher

January 2026 – Present

SLAC National Accelerator Laboratory

Palo Alto, CA

- Assisted in beam operations and developed a GUI using the EPIC system to increase speed and reliability for Schottky scans.
- Created a model and ran fluid simulations using Ansys Fluent for a plasma capillary to be used in the beamline.

ESP Associates Geotechnical Engineering

July 2025 – September 2025

ESP Associates

Fort Mill, SC

- Assisted with and led field and laboratory testing of various materials, including soils and concretes.
- Drilled, observed, and tested boreholes for ground water before building development.
- Obtained nuclear gauge certification, requiring radiation training from the NRC and DOT.

Research Assistant

August 2023 – March 2024

Duke University

Durham, NC

- Developed Python application to characterize carbon nanotube field emission sources for use in miniature mass spectrometers and microfocus x-rays. Assisted in the growth of carbon nanotubes using CVD.
- Gained experience utilizing lab equipment including Vacuum Chambers and Sourcemeters.

SteelFab Structural Engineering Intern

June 2023 – August 2023

SteelFab

Charlotte, NC

- Reviewed DRIs and DCPs, checking for faults and failures within the connections using excel and provided in-house software. Designed bolted and welded connections to prevent design failures
- Developed experience working with in-house software to model steel beams and connections.
- Visited and inspected construction sites and steel workshops to check connections and procedures

PROJECTS

PointBlank

- Currently developing an automatic targeting system to defend against drones and personnel combat.
- Helped develop a YOLOv8 model for target detection. Designed and 3D printed the frame of the scope, along with functionality updates for live feeds.

Curling Robot

- Developed a robot that could effectively score and play the Olympic sport of curling
- Built the power systems for the robot and assisted group mates with material acquisition, laser cutting of the frame, programming, and partitioning of work.
- Robot got second place in curling competition at Stanford.

RELEVANT SKILLS

Programming and Modeling

- Created a Python GUI to model IV step in Carbon nanotubes and improve data transfer for labs.
- General experience utilizing MATLAB in the classroom and for projects.
- Experience using CAD including Autodesk Inventor, Fusion 360, SteelFab in-house modeling, and Ansys Fluent.

Data Science

- Utilized Excel and Stata for data analysis in classes and at while developing connections for SteelFab
- Experience effectively communicating collected data, through verbal or visual sources.